# Phase I – Business Process Explanation

Rwanda Food Price Watch · Bruce RUTAGENGWA (28468) · INSY 8311 – Database Development with PL/SQL · Group C · Instructor: Eric Maniraguha

### Selected process: monthly market price collection, validation and spike alert

This process turns a price written down in a Rwandan market into a stored, checked price observation, and, when the price rises abnormally, into an alert that a district officer acts on. It addresses the problem stated in Phase 0: food price spikes are noticed late because prices only exist in a flat file. Its objectives are complete and correct monthly prices for every market and commodity, automatic detection of increases above an agreed threshold, and a recorded response to every confirmed spike. The expected outcome is an early warning that reaches the people who can act, together with a full audit trail.

### How the process begins and progresses

The process starts on a timer: on the 15th of each month the **Market Price Enumerator** visits the assigned market, records the retail or wholesale price of each commodity and unit, and hands the market price sheet to the **Data Entry Officer**. The officer enters the prices one by one through the *record_price* procedure. Inside *record_price* the **Price Watch System** first validates that the market, commodity and unit exist and that the price is greater than zero. An invalid value goes back to the officer, who checks it with the enumerator and corrects the sheet. A valid value is then inserted, which fires the restriction trigger: on a day when entry is not allowed the insert is blocked, the denied attempt is written to AUDIT_LOG, and the officer re-enters on an allowed day. Otherwise the price is saved in PRICE_OBSERVATION and the trigger *trg_price_audit* logs the insert. The loop repeats until the sheet is complete.

At 02:00 each night a DBMS_SCHEDULER job compares, for each market and commodity, the latest price with the previous month and with the commodity's limit in PRICE_THRESHOLD. When the increase is above the limit it creates a PRICE_ALERT with status OPEN; in all cases the Power BI dashboard data is refreshed.

### What each actor does and how decisions are made

The **Food Security Analyst** reviews open alerts on the dashboard and decides whether the spike is genuine. If it comes from a wrong entry, the alert is marked FALSE_ALARM and the Data Entry Officer corrects the price (an UPDATE that the audit trigger also logs). If it is genuine, the alert is confirmed and an alert notice is sent to the **District Agriculture Officer**, who assesses the market and decides whether intervention is needed, for example informing cooperatives or requesting a stock release, and then sends an action report. The analyst records the action and closes the alert, and closed alerts older than 12 months are archived. The system takes the rule-based decisions (validity, entry day, threshold); people take the judgement decisions (genuine spike, intervention).

### How information moves

Information moves on paper or mobile form from the enumerator to the data entry officer (market price sheet), from the officer into Oracle through a procedure, inside Oracle from observations to alerts, from Oracle to Power BI for the analyst, and finally as an alert notice to the district and an action report back. Every create, update and archive step is stored, so each alert can be traced back to the observations and the people behind it.

### Support for operations and management decisions

For daily operations the process guarantees that only valid prices on allowed days enter the database, and that every change is audited. For management it gives district and national decision-makers a monthly, map-based view of where staple prices are rising, a list of open and confirmed alerts, and a history of how quickly each one was handled, so that market interventions can be targeted and timed rather than reactive.

# Supporting tables

Diagram: business_process_diagram.pdf (BPMN 2.0, drawn in draw.io)

### Actors and responsibilities

| Actor | Type | Responsibility |
|---|---|---|
| Market Price Enumerator | Field employee (WFP / MINAGRI partner) | Visits the market on collection day, records each commodity price and unit, submits the price sheet, clarifies doubtful values. |
| Data Entry Officer | Employee | Enters prices through *record_price*, re-enters blocked entries on allowed days, corrects invalid or wrong prices. |
| Price Watch System (Oracle DB + PL/SQL) | System | Enforces the restriction rule, validates and stores prices, audits every change, runs the nightly spike detection, creates alerts, archives old alerts. |
| Power BI dashboard | System (innovation) | Shows prices, trends and open alerts to the analyst, connected live to Oracle. |
| Food Security Analyst | Employee (food security unit) | Reviews alerts, separates data errors from real spikes, notifies the district, records the action and closes alerts. |
| District Agriculture Officer | External stakeholder (local government) | Assesses the affected market, decides on intervention, informs cooperatives or requests stock release, reports back. |
| Cooperatives, traders and households | External stakeholders | Receive price information and benefit from earlier action; they do not enter data. |

### Data-generating activities

| Op | Activity (lane) | Table(s) | Information generated or used |
|---|---|---|---|
| R | Validate market, commodity, unit and price (System) | MARKET, COMMODITY, UNIT | Confirms the reference values exist before a price is accepted. |
| C | Block entry and log denied attempt (System) | AUDIT_LOG | User, time, attempted action and reason it was denied. |
| C | Save price observation (System) | PRICE_OBSERVATION | Market, commodity, unit, price type, month and price in RWF. |
| C | trg_price_audit logs the insert / update (System) | AUDIT_LOG | Old and new values, user and timestamp of each change. |
| R | Compute % change vs previous month (System) | PRICE_OBSERVATION, PRICE_THRESHOLD | Current and previous price, allowed % increase per commodity. |
| C | Create price alert (System) | PRICE_ALERT | Market, commodity, month, % change, threshold, status OPEN. |
| R | Refresh dashboard data (System) / Review open alerts (Analyst) | PRICE_OBSERVATION, PRICE_ALERT | Price trends by market and district; list of open alerts. |
| U | Mark alert FALSE_ALARM (Analyst) | PRICE_ALERT | Status changed to FALSE_ALARM with the reason. |
| U | Correct the wrong price (Data Entry Officer) | PRICE_OBSERVATION | Corrected price; old value kept in AUDIT_LOG by the trigger. |
| U | Confirm alert and notify district (Analyst) | PRICE_ALERT | Status CONFIRMED, notification date. |
| U | Record action taken and close alert (Analyst) | PRICE_ALERT | Action description, closing date, status CLOSED. |
| D | Archive closed alerts older than 12 months (System) | PRICE_ALERT | Old closed alerts moved out of the active table to keep it small. |